

Investment Appraisal Techniques

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Investment



Finance



Dividends

Key Decisions

Purchase of land
and machinery

Long- term Investment

Capital Expenditure

- **Objective** - Building resources for the long run
- **Expectations** - High returns (ROI or ROA)
- **Risk** - Regular fluctuations

Inventory and
debtors

Short term Investment

Working Capital

- **Objective** - Smooth day-to-day operations
- **Expectations** - Financial stability
- **Risk** - Inflation and Purchasing power

Estimation of Cash Flows

- **Incremental Cash Flow**

Additional cash flow that arise with the acceptance of a project

- **Net Cash Flow = Cash Inflows - Cash Outflows**

- **Effects of Inflation**

- **Effects of Taxation**

Initial Investment

- Cost of acquiring
 - Cost of setting up the project
-

Operating Cash Flows

Net cash inflows from operations over the project's life.

Terminal Cash Flows

Cash flows at the end of the project's life, including salvage value.

Components of Incremental Cash Flows

Inflation

**General Inflation
vs
Specific Inflation**

Types of Inflation

Problem 1

A project requires an initial investment of ₹100,000 and is expected to generate additional revenues of ₹30,000 per year with additional costs of ₹10,000 per year for 5 years. Assume an annual inflation rate of 3%.
Calculate the Incremental Cash Flows, adjusted for inflation.

Year	Cash Inflow	Cash Outflow	Adjusted Inflow	Adjusted Outflow	Incremental Cash Flow
0	-	100,000	-	100,000	-100,000
1	30,000	10,000	30,900	10,300	20,600
2	30,000	10,000	31,827	10,609	21,218
3	30,000	10,000	32,782	10,927	21,855
4	30,000	10,000	33,765	11,255	22,510
5	30,000	10,000	34,778	11,592	23,186

Problem 2

A company is evaluating a project with an initial investment of ₹60,000. The project is expected to generate revenues of ₹15,000 in the first year and grow by 4% annually due to inflation. The costs are ₹5,000 in the first year and grow by 2% annually due to inflation. The project lasts for 4 years.

Calculate the Incremental Cash Flows, adjusted for inflation.

Year	Cash Inflow	Cash Outflow	Incremental Cash Flow
0	-	60,000	-60,000
1	15,000	5,000	10,000
2	15,600	5,100	10,500
3	16,224	5,202	11,022
4	16,873	5,306	11,567

Relevant Cash Flows

- future
 - incremental
-

Discounting - DCF techniques

- PV of a single sum
 - PV of annuity
 - PV of perpetuity
-

Considerations

- Inflation
 - Taxation
 - Working Capital
-

Net Present Value (NPV)

$$NPV = \sum \left(\frac{C_t}{(1+r)^t} \right) - \text{Initial Investment}$$

Problem 3

Calculate the NPV for a project with the following cash flows, considering a discount rate of 10%:

- Initial investment: \$100,000
- Year 1: \$30,000
- Year 2: \$40,000
- Year 3: \$50,000

$$NPV = \sum \left(\frac{C_t}{(1 + r)^t} \right) - \text{Initial Investment}$$

	T0	T1	T2	T3
Cash Flow		30,000	40,000	50,000
Initial Investment	-100,000			
Present Value (PV)	-100,000	27,273	33,058	37,566

NPV = (27,273 + 33,058 + 37,566) - 100,000

NPV = -2,103

Problem 4

A company is considering a new project that requires an initial investment of \$50,000. The project is expected to generate additional revenues of \$30,000 per year and additional costs of \$5,000 per year for the next 4 years. Calculate the **Net Present Value** for the project, considering a discount rate of 10%.

	T0	T1	T2	T3	T4
Cash Inflow		30,000	30,000	30,000	30,000
Cash Outflow	-50,000	-5,000	-5,000	-5,000	-5,000
Net Cash Flow	-50,000	25,000	25,000	25,000	25,000
PV	-50,000	22,727	20,661	18,783	17,075
NPV	29,246				

Money/Nominal Method

Step 1

Inflate each cash flow by its specific inflation rate

Money Flow

Step 2

Calculate net annual cash flow

Net Cash Flow

Step 3

Discount using nominal rate

Discounting

Dealing with Inflation

Real Method

Step 1

Do not inflate cash flows

Real Flows

Step 2

Calculate net annual cash flow

Net Cash Flow

Step 3

Discount using real rate

Discounting

Fisher Formula

$$(1 + i) = (1 + r)(1 + h)$$

OR

$$(1 + r) = (1 + i)/(1 + h)$$

r = real rate

i = nominal interest rate

h = general inflation rate

Problem 5

A company is evaluating a project with an initial investment of \$60,000. The project is expected to generate revenues of \$35,000 and operating expenses of \$5,000 annually. The inflation rate is 5%. The project lasts for 4 years. Calculate the NPV, considering a discount rate of 10%.

<u>Real Method</u>	T0	T1	T2	T3	T4
Revenue		35,000	35,000	35,000	35,000
Operating Exp.		-5,000	-5,000	-5,000	-5,000
Initial Investment	-60,000				
Net Cash Flow	-60,000	30,000	30,000	30,000	30,000
PV	-60,000	27,273	24,793	22,539	20,490
NPV	35,096				

Nominal Method

Cost of Capital:

$(1 + i) = (1 + r)(1 + h)$

$(1 + i) = (1 + 0.1)(1 + 0.05)$

$(1 + i) = 1.155$

$i = 0.155$ (or 15.5%)

	T0	T1	T2	T3	T4
Revenue		36,750	38,588	40,517	42,543
Operating Exp.		-5,250	-5,513	-5,788	-6,078
Initial Investment	-60,000				
Net Cash Flow	-60,000	31,500	33,075	34,729	36,465
PV	-60,000	27,273	24,793	22,539	20,490
NPV	35,096				

Problem 6

Calculate the NPV considering:

- Initial investment: \$150,000
- Year 1-4 cash inflows: \$45,000 annually
- Working capital requirement: 10% of annual inflows
- Inflation rate: 5%
- Discount rate: 12%

	T0	T1	T2	T3	T4
Cash Inflow		47,250	49,613	52,093	54,698
Difference		2,363	2,481	2,605	
Working Capital	$47,250 \times 0.1 = 4725$	$2,363 \times 0.1 = 236$	$2,481 \times 0.1 = 248$	$2,605 \times 0.1 = 260$	5,470 (Inflow)

Cost of Capital:

$(1 + i) = (1 + r)(1 + h)$

$(1 + i) = (1 + 0.12) (1 + 0.05)$

$(1 + i) = 1.176$

$i = 0.176 \text{ (or } 17.6\%)$

<u>Nominal Method</u>	T0	T1	T2	T3	T4
Cash Inflow		47,250	49,613	52,093	54,698
Initial Investment	-150,000				
Working Capital	-4725	-236	-248	-260	5,470
Net Cash Flow	-154,725	47,014	49,364	51,833	60,168
PV	-154,725	39,978	35,694	31,870	31,458
NPV	-15,725				

Real Method

- Discount rate: 12%

	T0	T1	T2	T3	T4
Cash Inflow		45,000	45,000	45,000	45,000
Initial Investment	-150,000				
Working Capital	-4,500				4,500
Net Cash Flow	-154,500	45,000	45,000	45,000	49,500
PV	-154,500	40,179	35,874	32,030	31,458
NPV	-14,959				

Nominal Method vs Real Method

- Real method can only be used if all cash flows are inflating at the general rate of inflation.
- In questions involving specific inflation rates, taxation or working capital, the nominal method is usually more reliable.

Problem 7

A company plans to invest \$5 million in new manufacturing equipment. The equipment is expected to increase revenues by \$2.5 million annually for 5 years. The annual maintenance cost is \$0.5 million, and wages will increase by \$0.3 million per year. Additional working capital of 12% of the annual revenue increase is required at the start of each year. Working capital will be released at the end of the project.

The inflation rates are as follows:

- Revenues: 8%
- Wages: 4%
- Maintenance: 5%
- General prices: 6.5%

The company's real cost of capital is 8%.

Calculate the net annual cash flows of the project and determine NPV.

Cost of Capital:

$(1 + i) = (1 + r)(1 + h)$

$(1 + i) = (1 + 0.08) (1 + 0.065)$

$(1 + i) = 1.15$

$i = 0.15 \text{ (or 15\%)}$

(in '000)	T0	T1	T2	T3	T4	T5
Revenue (Infl. @ 8%)		2,700	2,916	3,149	3,401	3,673
Difference		216	233	252	272	
Working Capital	2,700 x 0.12 = 324	216 x 0.12 = 26	233 x 0.12 = 28	252 x 0.12 = 30	272 x 0.12 = 33	441 (Inflow)

(in '000)	T0	T1	T2	T3	T4	T5
Revenue (Infl. @ 8%)		2,700	2,916	3,149	3,401	3,673
Wage Cost (Infl. @ 4%)		-312	-324	-337	-351	-365
Operating Cash Flow		2,388	2,592	2,812	3,050	3,308
Investment	-5,000					
Maintenance (Infl. @ 5%)		-525	-551	-579	-608	-638
Working Capital	-324	-26	-28	-30	-33	441
Net Cash Flow	-5,324	1,837	2,012	2,203	2,410	3,111
PV Factor @ 15%	1.000	0.870	0.756	0.658	0.572	0.497
PV	-5,324	1,598	1,521	1,450	1,379	1,546
NPV	2,170					

Impact of Taxation

Charged on Operating Cash Flows

- Operating cash inflows will be taxed
- Operating cash outflows will be tax deductible

Tax Allowable Depreciation (TAD)

- Capital allowances or writing down allowances
- Generates tax relief
- TAD can be calculated on Straight-Line (SLN) or Written Down Value (WDV) basis.
- TAD is claimed as early as possible.
- In the year of sale or scrap, a balancing allowance or charge arises

Problem 8

A company invests \$100,000 in equipment with a useful life of 4 years and salvage value of \$5000. It will be depreciable on WDV basis at 30%. The project is expected to generate additional revenues of \$40,000 per year and additional costs of \$10,000 per year. The corporate tax rate is 30%. Ignore inflation.

Calculate the NPV, considering a discount rate of 10%.

Initial Investment = 100,000

Salvage value = 5,000

Useful life = 4 years

Depreciation - WDV basis @ 30%

TAD calculation:

Initial Investment	100,000
TAD Year 1	$= 100,000 \times 0.3 = \mathbf{30,000}$
TAD Year 2	$= (100,000 - 30,000) \times 0.3 = \mathbf{21,000}$
TAD Year 3	$= (100,000 - 30,000 - 21,000) \times 0.3 = \mathbf{14,700}$
Salvage Value	5,000
TAD Year 4	$100,000 - 30,000 - 21,000 - 14,700 - 5,000 = \mathbf{29,300}$

	T0	T1	T2	T3	T4
Revenue		40,000	40,000	40,000	40,000
Cost		-10,000	-10,000	-10,000	-10,000
TAD		-30,000	-21,000	-14,700	-29,300
Taxable Profit		0	9,000	15,300	700
Tax @ 30%		0	-2,700	-4,590	-210
Add back TAD		30,000	21,000	14,700	29,300
Salvage value					5,000
Investment	-100,000				
Net Cash Flow	-100,000	30,000	27,300	25,410	34,790
PV	-100,000	27,270	22,550	19,083	23,762
NPV	-7,312				

Problem 9

An inventory management system will cost Rs.1crore. It will be depreciable on SLN basis for 5 years which is its useful life. At the end of 5th year, it can be disposed for Rs.20 lakhs. This system will annually save Rs.30 lakhs before depreciation and tax. The tax rate is 30% and the system will require Rs. 10 lakhs initially as working capital. Assuming 10% cost of capital, calculate the net investment, net annual cash flows, terminal cashflow & NPV of this project.

Initial Investment = 1,00,00,000

Salvage value = 20,00,000

Useful life = 5 years

Depreciation - SLN basis

$TAD = (100,00,000 - 20,00,000)/5$

TAD = 16,00,000

(in '000)	T0	T1	T2	T3	T4	T5
Revenue		3,000	3,000	3,000	3,000	3,000
TAD		-1,600	-1,600	-1,600	-1,600	-1,600
Taxable Profit		1,400	1,400	1,400	1,400	1,400
Tax		-420	-420	-420	-420	-420
Add back TAD		1,600	1,600	1,600	1,600	1,600
Scrap Value						2,000
Working Capital	-1,000					1,000
Investment	-10,000					
Net Cash Flow	-11,000	2,580	2,580	2,580	2,580	5,580
PV	-11,000	2,345	2,132	1,938	1,762	3,465
NPV	643					

Problem 11

Avighna Technologies, a mid-sized industrial robotics firm, is evaluating a proposal to launch a new line of automated material-handling equipment. The project has a life of 4 years and requires an initial investment in machinery costing ₹40,00,000. The machinery will be depreciated using the straight-line method over 4 years, with no salvage value. The company expects to sell 1,000 units annually at a price of ₹12,000 per unit in the first year, with selling prices expected to grow at a general inflation rate of 4% per annum. The variable cost per unit is ₹6,000, expected to grow at a specific inflation rate of 6% per annum due to input cost pressures. Fixed operating costs (excluding depreciation) are ₹10,00,000 in Year 1 and are also expected to increase by general inflation of 4% annually.

To support operations, the project will require net working capital investment equal to 10% of the projected sales revenue each year. The working capital is to be invested at the beginning of each year and will be fully recovered at the end of the fourth year. The company's tax rate is 30%, and the project's nominal required rate of return is 14% (including general inflation). All cash flows occur at year-end unless otherwise stated.

Calculate the Net Present Value (NPV) of the project. Should Avighna Technologies undertake the project?

Internal Rate of Return (IRR)

Discount rate at which the $NPV = 0$

Found by linear interpolation

If $IRR > \text{Cost of Capital}$, ACCEPT the project
If $IRR < \text{Cost of Capital}$, REJECT the project

Provides information about the riskiness of a project

Positive NPV only when the firm's
 $\text{Cost of Capital} < IRR$

Step 1

Calculate two NPVs for the project at two different costs of capital.

Step 2

Calculate IRR

$$\text{IRR} = L + \frac{N_L}{(N_L - N_H)} \times (H - L)$$

where:

L = Lower rate of interest.

H = Higher rate of interest.

NL = NPV at lower rate of interest.

NH = NPV at higher rate of interest.

Step 3

Compare the IRR with the company's cost of borrowing.

Problem 12

EcoGreen Technologies, a company focused on sustainable innovations, is considering an initial investment of \$2,500 in a new project. The project is expected to generate cash inflows of \$600, \$600, \$700, \$700, and \$820 at 12-month intervals. The project does not have any residual value. The required rate of return for financing the project is 9%. **Calculate the IRR.**

	Cash Inflow	PV @ 8%	PV @ 12%
T0	-2,500	-2,500	-2,500
T1	600	556	536
T2	600	514	478
T3	700	556	498
T4	700	515	445
T5	820	558	465
NPV		198	-78

$$IRR = L + \frac{N_L}{(N_L - N_H)} \times (H - L)$$

$$IRR = 0.08 + \frac{198}{(198 - (-78))} \times (0.12 - 0.08)$$

$$IRR = 0.109 \text{ (or 10.9\%)}$$

Assumptions

Funds can be reinvested at the IRR for the duration of the project

Decision rule is not always clear cut

In case of multiple IRRs, difficult to compare with Cost of Capital

Choosing between projects

Difficult to compare projects as projects can have multiple IRRs

Problems with using IRR

Modified IRR (MIRR)

Measure of the return from a project

MIRR = Project's return

Formula:

$$MIRR = \left[\frac{PVR}{PVI} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

PVR = the present value of the 'return phase' of the project

PVI = the present value of the 'investment phase' of the project

re = the firm's reinvestment rate

Discounted Payback Period (DPP)

Payback period does not account TVM

DPP measures the length of time before the discounted cash returns from a project cover the initial investment.

Shorter the discounted payback period, the more attractive the project

Long discounted payback period indicates that the project is a high risk project.

Problem 13

A project with the following cash flows is under consideration:

- Initial investment: \$2.5 million
- Year 1: 1.1 million
- Year 2: 860,000
- Year 3: 700,000
- Year 4: 720,000
- Year 5: 820,000

Assuming 9% cost of capital, calculate the **Discounted Payback Period**.

(in 000s)	Cash flow	Discounted Cash Flow	Cumulative Discounted Cash Flow
T0	-2500	-2500	-2500
T1	1100	1009	-1491
T2	860	724	-767
T3	700	541	-226
T4	720	510	284
T5	820	533	817

$$\text{Discounted Payback Period} = 3 + (226/510)$$

$$\text{Discounted Payback Period} = 3 + 0.44$$

$$\text{Discounted Payback Period} = 3.44$$

Profitability Index

Return for each dollar/rupee invested

$$PI = \frac{\text{PV of future cash flows}}{\text{Initial investment}}$$

OR

$$PI = 1 + \frac{NPV}{\text{Initial investment}}$$

If $PI > 1$, ACCEPT the project

If $PI < 1$, REJECT the project

Problem 14

A project with the following cash flows is under consideration:

- Initial investment: \$2.5 million
- Year 1: 1.1 million
- Year 2: 860,000
- Year 3: 700,000
- Year 4: 720,000
- Year 5: 820,000

Assuming 9% cost of capital, calculate the **Profitability Index**.

(in 000s)	Cash flow	Discounted Cash Flow
T0	-2500	-2500
T1	1100	1009
T2	860	724
T3	700	541
T4	720	510
T5	820	533

$$PI = \frac{\text{PV of future cash flows}}{\text{Initial investment}}$$

$$PI = \frac{1,009 + 724 + 541 + 510 + 533}{2,500}$$

$$PI = \frac{3,317}{2,500} = 1.33$$

Capital Rationing

Allocation of scarce financial resources

Objective is to achieve optimal returns

Types of Capital Rationing:

Hard capital rationing - External factors

- Difficult to raise additional capital
- Poor track record
- Low credit rating

Soft capital rationing - Internal policies

- Maintain financial ratios
 - Encourages acceptance of only substantially profitable business
-

Dealing with
Capital Rationing
- **Profitability
Index**

Single Period

- **Divisible Projects**
- **Indivisible Projects**
- **Mutually Exclusive Projects**

Dealing with
Capital Rationing
- **Linear
Programming**

Multi Period

Objectives:

- maximise NPV from the investment in available projects
- maximise the PV of cash flow available for dividends

Problem 15

FutureTech Innovations is a company specialising in cutting-edge technology solutions. The company can only raise \$1 million of finance to invest at T0. The management team has identified five potential projects, each with its own initial investment requirement, NPV, IRR, and discounted payback period. The company's cost of capital is 10%. Below are the details of the projects:

Project	Initial Investment (\$ 000)	NPV (\$ 000)	IRR (%)	Discounted Payback Period (years)
A	300	50	14	4.5
B	200	40	15	3.8
C	500	70	12	5.2
D	400	60	13	4.9
E	250	45	16	3.5

Advise the company which project(s) to accept if the projects are:

- 1.independent and divisible
- 2.independent and indivisible
- 3.mutually exclusive

Funds - \$1 million

Cost of capital - 10%

Project	Initial Investment (\$ 000)	NPV (\$ 000)	IRR (%)	Discounted Payback Period (years)	Profitability Index (PI)
A	300	50	14	4.5	1.17
B	200	40	15	3.8	1.20
C	500	70	12	5.2	1.14
D	400	60	13	4.9	1.15
E	250	45	16	3.5	1.18

1. Independent and divisible

Consider PI method to find the best possible combination:

- First Project B (\$200,000), then Project E (\$250,000), and Project A (\$300,000)
- At last partial Project D (62.5%)

2. Independent and indivisible

Possible choices:

- A, B, C ($\text{NPV} = 50 + 40 + 70 = 160$)
- B, C, E ($\text{NPV} = 40 + 70 + 45 = 155$)
- A, B, D ($\text{NPV} = 50 + 40 + 60 = 150$)
- A, B, E ($\text{NPV} = 50 + 40 + 45 = 135$)
- C, D ($\text{NPV} = 70 + 60 = 130$)

3. Mutually exclusive

Select the one with the highest NPV: Project C

Project	Initial Investment (\$ 000)	NPV (\$ 000)	IRR (%)	PI
A	300	50	14	1.17
B	200	40	15	1.20
C	500	70	12	1.14
D	400	60	13	1.15
E	250	45	16	1.18

